

ANDY NGUYEN

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Education

University of California, Irvine

Sep. 2021 – Jun. 2025

Bachelor of Science in Computer Science

Irvine, CA

- GPA: 3.8
- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming (Python), Databases, Information Retrieval, Computer Organization, Software Engineering Principles, Computer Networks

Experience

Student Software Engineer - Capstone Project

Jan. 2025 – Jun. 2025

Sound Ethics

Los Angeles, CA

- Developed deep learning models to distinguish between authentic and AI-generated audio to protect artists' rights from generative music.
- Investigated and built on current audio deepfake detection research and techniques, including CNNs, spectrogram analysis, and residual learning.
- Achieved 85% accuracy on augmented audio using a RawNet with a Cosine Classifier and contrastive learning, improving by 40%.

UCI Undergraduate Reader

Mar. 2025 – Jun. 2025

UCI Department of Informatics

Irvine, CA

- Grading coursework for 200+ students in Information Retrieval, providing timely feedback to support academic growth.

UCI Computer Science Lab Tutor

Sep. 2022 – Dec. 2023

UCI School of ICS

Irvine, CA

- Mentored over 50 students weekly in introductory Python courses, simplifying complex concepts with concise example-driven guides and debugging assistance.

Projects

Dungeon Scribe | *Python, AWS Bedrock, AWS Lambda, React*

- Developed an AI assistant that supports Dungeons & Dragons players with character creation and generates immersive, character-based roleplay guidance using AWS Bedrock (Mistral Large 2).
- Built a serverless backend using Python, AWS Lambda, and API Gateway to provide quick feedback from structured LLM prompts.
- Evaluated multiple AWS Bedrock models and refined prompt formatting to produce consistent, high-quality, and context-aware responses.

Fabflix | *Java, MySQL, AWS EC2, React*

- Developed a full-stack movie database on AWS EC2, handling 90,000+ movie records with efficient data retrieval.
- Designed backend APIs with Java web servlets and JDBC for secure data retrieval and HttpSession for persistent user cart management.
- Enhanced performance with load balancing for high traffic and Kubernetes deployment with Docker for scalability.

Fashion-MNIST Classifier Investigation | *Python, Scikit-learn, PyTorch Matplotlib, NumPy, Jupyter Notebook*

- Evaluated 4 machine learning classifiers on 70,000 Fashion-MNIST images.
- Trained and tuned hyperparameters to achieve 85% accuracy with Logistic Regression, K-Nearest Neighbors, and Feed-Forward Neural Network classifiers.
- Optimized and achieved 99% accuracy with a Convolutional Neural Network when predicting the clothing type.

Technical Skills

Languages: Python, Java, C, C++, HTML/CSS, SQL

Technologies/Frameworks: Amazon Web Services, Google Cloud Platform, React, JUnit, AJAX, jQuery, MaterialUI, MySQL, PostgreSQL, PyTorch, Scikit-learn, NumPy

Developer Tools: Docker, Git, VS Code, Eclipse, Jupyter Notebook

Leadership

Technology Chair | Circle K International at UCI

Mar. 2024 – Feb. 2025

- Managed club website for over 150 members, delivering updated club information and news.
- Maintained MySQL database storing event details and member involvement.

Membership Programs Chair | Circle K International at UCI

Mar. 2024 – Feb. 2025

- Prepared 15 members for leadership positions to create welcoming and cohesive family groups and host large-scale programs for general club members.